FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY (FISAT)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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ACADEMIC YEAR : 2026 – 2027 **BATCH :** CSD

Seminar Title :	**Demystifying Defects: Federated Learning and Explainable AI for Semiconductor Fault Detection**

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Type of Seminar :

[✓] **Review Seminar** (presenting a journal/conference/survey paper)

[ ] Technical / Concept Seminar (explaining an algorithm, technology, architecture, protocol, etc.)

[ ] Application Seminar (showing how a technology is applied in real-world systems)

[ ] Product / Tool Seminar (overview of a software/hardware product, platform, or framework)

Project Mapping

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
x	x	x	x	x	x	x	x	x	x	x	x	x

Mapping Justification

S No	PO/PSO	Justification
1	PO1	Apply computer science and engineering knowledge to understand distributed machine learning, convolutional neural networks, federated averaging, and gradient optimization.
2	PO2	Analyze semiconductor wafer defect patterns (center, donut, edge-ring, mixed types) and examine the privacy limitations of centralized training across proprietary fabrication facilities.
3	PO3	Understand the design of a privacy-preserving federated inspection framework combining ResNet152 backbones, FedAvg aggregation, differential privacy, and explainable AI.
4	PO4	Examine the benchmark dataset (MixedWM38 with 38,000+ images), exploratory data analysis, PCA dimensionality reduction, and evaluation metrics across 10 communication rounds.

S No	PO/PSO	Justification
5	PO5	Understand the use of PyTorch, differential privacy libraries, Grad-CAM, LIME, and t-SNE visualization tools in implementing the semiconductor fault detection framework.
6	PO6	Recognize how automated defect detection in semiconductor manufacturing enhances hardware quality, device reliability, and cost efficiency in automotive and medical electronics.
7	PO7	Understand the importance of data confidentiality in industrial AI, safeguarding proprietary wafer map designs through differential privacy without exposing IP.
8	PO8	Develop the ability to independently study peer-reviewed IEEE Access research, organize technical findings, and collaborate in preparing and presenting the seminar.
9	PO9	Improve technical communication by clearly explaining federated aggregation formulas, Gaussian noise injection, Grad-CAM heatmaps, and performance benchmarks.
10	PO10	Understand resource planning, communication round budgets, and computational requirements when deploying federated learning models across distributed manufacturing fabs.
11	PO11	Develop the ability to explore emerging frontiers in edge federated learning, MLOps CI/CD pipelines, and quantum-accelerated semiconductor inspection.
12	PSO1	Apply deep learning and image processing principles to model semiconductor wafer map patterns and design explainable visual feature attribution mechanisms.
13	PSO2	Understand the integration of privacy-preserving decentralized algorithms with post-hoc interpretability models for reliable industrial AI engineering.

Signature (Seminar Coordinator) : _____

ABSTRACT

Semiconductor manufacturing underpins nearly every modern electronic device, from mobile consumer electronics to mission-critical automotive and healthcare systems. Detecting wafer defects early during fabrication is vital to maximizing die yield, preventing downstream failures, and curtailing manufacturing overhead. However, conventional machine learning defect detection systems rely heavily on centralized data collection, requiring original equipment manufacturers (OEMs) and foundries to share proprietary wafer inspection maps—a practice strongly resisted due to intellectual property and trade secret concerns. Furthermore, state-of-the-art deep neural networks operate as opaque black boxes, providing negligible insight to process engineers as to why specific wafer patterns are flagged defective. This seminar presents a detailed review of the paper **"Demystifying Defects: Federated Learning and Explainable AI for Semiconductor Fault Detection"** by T. Patel, R. Murugan, G. Yenduri, R. H. Jhaveri, H. Snoussi, and T. Gaber, published in *IEEE Access*, Vol. 12, 2024. The paper introduces a privacy-preserving, transparent inspection framework combining decentralized federated training with post-hoc explainability.

The proposed framework distributes convolutional neural network backbones across ten client manufacturing stations, training models locally on private wafer image shards while ensuring raw data never leaves client boundaries. A central coordinator aggregates model weights via Federated Averaging (FedAvg) enhanced with calibrated Gaussian noise to enforce formal Differential Privacy against reconstruction attacks. Four deep learning architectures (ResNet152, DenseNet121, MobileNetV2, InceptionV3) were evaluated on the benchmark MixedWM38 dataset comprising over 38,000 grayscale wafer maps across 38 defect classes. The ResNet152 federated model achieves a peak test accuracy of **98.78%**, outperforming prior deformable convolutional network baselines (DCNet, 93.20%) by nearly 5.6 percentage points. An integrated Explainable AI layer combining Gradient-weighted Class Activation Mapping (Grad-CAM) and Local Interpretable Model-agnostic Explanations (LIME) produces interpretable heatmaps and super-pixel attributions that confirm the model focuses on genuine physical defect clusters rather than background noise. This approach demonstrates that high detection accuracy, privacy preservation, and model transparency can be simultaneously achieved in modern semiconductor fabrication.

Keywords : Semiconductor Manufacturing, Wafer Defect Detection, Federated Learning, Explainable AI (XAI), Differential Privacy, Grad-CAM, LIME.

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3.3 Study of Proposed System The proposed system combines federated learning with explainable artificial intelligence for wafer defect detection. Rather than relying on centralized data pooling that forces semiconductor foundries to expose proprietary fabrication data, the framework trains a global convolutional neural network across distributed clients without moving raw wafer images. The framework uses data collected through the MixedWM38 wafer map benchmark comprising over 38,000 grayscale wafer images across 38 fault classes. The study examines wafer spatial patterns Department of Computer Science and Design,FISAT 10Demystifying Defects: Federated Learning and Explainable AI for Semiconductor Fault Detection characteristics such as center, donut, edge-ring, and mixed-type defect clusters. These characteristics are used to assess whether individual dies across the wafer satisfy strict semiconductor manufacturing yield standards. The collected wafer information is incorporated into the federated aggregation process. This enables the central model to learn robust representations across non-IID client distributions while preserving facility data confidentiality. The model first trains localized convolutional feature extractors across ten manufacturing nodes. When local client epochs conclude, the system applies Federated Averaging with calibrated Gaussian noise injection to provide differential privacy guarantees. The paper also presents an explainable AI layer combining Grad-CAM and LIME to support the practical auditing of code. The application is intended to communicate feature attribution heatmaps directly to foundry process engineers, making defect detection trustworthy and actionable. The overall approach connects decentralized training, privacy-preserving aggregation, and visual interpretability in a single framework. 3.4 Requirement Specification The proposed semiconductor fault detection framework involves wafer image ingestion, local client model training, secure gradient aggregation, and post-hoc visual explanation generation. Therefore, suitable data, processing capabilities, and software components are required to support the workflow. The study depends on the availability of standardized wafer map data derived from multi-stage fabrication sensors. The collected data must describe thirty-eight distinct single and mixed-type wafer defect patterns. A suitable computational environment is also required to process deep convolutional backbones and calculate differential privacy budgets. 3.4.1 Recommended System Requirements The following requirements describe the general computational resources needed to support a federated wafer inspection framework. They should not be interpreted as exact hardware specifications reported by the paper: • Processor: A high-performance multi-core CPU coupled with dedicated GPU acceleration (NVIDIA RTX 3080 or equivalent) to handle parallel convolutional tensor calculations. • Memory: Sufficient RAM to store normalized 52x52 wafer batch arrays and gradient update caches (Minimum 16GB, 32GB recommended). • Storage: Adequate high-speed SSD storage for caching the MixedWM38 dataset, model checkpoints, and t-SNE clustering projections. • Network Connectivity: Low-latency local network or TLS-encrypted tunnel required for transferring weight update tensors between client inspection nodes and the central aggregation server. • Execution Environment: Linux-based operating system (Ubuntu 22.04 LTS) configured with CUDA acceleration libraries for client training. Department of Computer Science and Design,FISAT 11Demystifying Defects: Federated Learning and Explainable AI for Semiconductor Fault Detection • Explainability Module: Dedicated compute capacity to generate pixel-level Grad-CAM heatmaps and LIME super-pixel perturbation surrogates for real-time defect verification. 3.4.2 Software Requirements The framework requires software components capable of training deep residual networks, managing decentralized communication rounds, injecting differential privacy noise, and generating visual heatmaps: • Data Preprocessing: Image processing libraries for grayscale normalization, pixel standardization, and one-hot encoding of 38 defect categories. • Machine Learning Engine: PyTorch / TensorFlow runtime supporting ResNet152, DenseNet121, MobileNetV2, and InceptionV3 backbones. • Federated Orchestration: Framework supporting Federated Averaging (FedAvg) and client node scheduling across 10 communication rounds. • Privacy Engine: Differential privacy routines for calibrated Gaussian perturbation of global weight updates. • Explainable AI Suite: Grad-CAM gradient backpropagation modules and LIME local surrogate algorithms for interpretable inspection. 3.5 Functional Requirements The federated wafer fault detection framework should satisfy the following functional requirements: 1. The system shall partition the wafer map dataset into ten private client shards representing decentralized fabrication stations. 2. The system shall train local convolutional models on private wafer image data without exporting raw wafer images off-site. 3. The system shall evaluate four deep learning architectures (ResNet152, DenseNet121, MobileNetV2, InceptionV3) using residual skip connections. 4. The system shall combine client weight updates at a central coordinator server using Federated Averaging (FedAvg). 5. The system shall inject calibrated Gaussian noise into aggregated model weights to guarantee formal differential privacy. 6. The system shall achieve high classification accuracy across all thirty-eight single, two-mixed, three-mixed, and four-mixed defect categories. 7. The system shall generate Grad-CAM class activation heatmaps highlighting the exact die regions responsible for defect classification. Department of Computer Science and Design,FISAT 12Demystifying Defects: Federated Learning and Explainable AI for Semiconductor Fault Detection 8. The system shall construct LIME super-pixel segmentation overlays to corroborate model focus. 9. The system shall support feature space visualization via t-SNE clustering to verify cross-client latent representation quality. 10. The system shall support the presentation of automated defect inspection

recommendations through an intuitive inspection dashboard for fab engineers. 3.6 Non-Functional Requirements The federated semiconductor inspection framework should satisfy the following non-functional requirements: • Privacy Preservation: Raw wafer images must strictly reside within client fabrication boundaries, and differential privacy noise must prevent gradient reconstruction attacks. • Accuracy and Reliability: The global federated classifier must achieve high generalization accuracy, matching or exceeding centralized baselines (reaching a verified 98.78% test accuracy with ResNet152, outperforming DCNet by 5.58 percentage points). • Convergence and Communication Efficiency: The federated optimization protocol must achieve stable convergence within a constrained budget of 10 communication rounds. • Transparency and Explainability: Model predictions must be accompanied by faithful, human-interpretable visual explanations (Grad-CAM and LIME) to eliminate black-box opacity and establish domain trust among foundry engineers. • Robustness to Data Heterogeneity: The framework must maintain stable classification performance across unbalanced defect distributions and varying client dataset volumes. • Scalability: The architecture must easily scale to accommodate additional manufacturing facilities and new wafer geometry formats without requiring architectural reconfiguration. • Maintainability: Modular separation between the client training nodes, aggregation server, and XAI interpretation pipeline must allow independent algorithmic updates. 3.7 Summary This chapter analysed the system overview, architectural design, dataset characteristics, and technical requirements of the federated learning and explainable AI framework for semiconductor defect detection. The reviewed framework addresses the privacy bottlenecks of centralized training and the black-box opacity of deep neural networks by combining decentralized FedAvg, differential privacy, and dual XAI interpretability layers. The proposed approach connects private localized training on 38,000 wafer maps with transparent attribution heatmaps, demonstrating superior accuracy and operational trustworthiness for multi-stakeholder industrial semiconductor manufacturing. Department of Computer Science and Design,FISAT